

Portable Air Quality Monitor

Category: Environmental Monitoring | Status: Completed Prototype | 7 min read

An Arduino air quality monitoring prototype using DHT11 and MQ sensors to explore gas sensing, calibration, and environmental awareness.

Materials

- Arduino UNO
- USB cable
- Breadboard
- Jumper wires
- DHT11 sensor
- MQ-3 sensor
- MQ-9 sensor
- MQ-135 sensor

What I learned

- Air quality data needs careful interpretation.
- Sensor arrays can provide a broader picture than one sensor alone.
- Environmental health projects need honesty about accuracy limits.

Safety or accuracy note

Low-cost sensors need calibration and should be treated as approximate indicators, not certified measurements.